np.random.seed (42)
↓
every time read same random numbers

- randint (
- normal (

Using Sklearn
have datasets for regression & classification
load_iris : read from sklearn
iris.data ; view feature columns
iris.target; view target column.

mean
std
readcsv
read_json
txt

Label Encoding
One - Hot Encoding

→ sklearn.preprocessing import labelEncoder
→ sklearn.preprocessing import one Hot Encoder.
↳ binary form

1. Data Preprocessing
↳ data cleaning

① Missing values :
replace missing value with mean of whole column
df.dropna (axis = 0 → row
df.fillna (df.median())
Handling

② Catagorical Values
Ordinal Encoding

③ Handling Outliers
IQR

Q1 = data[ ].quatile(0.25)
Q_3 = data .quatile(0.75)

IQR = Q3 - Q1

lower limit = Q1 - 1.5 * IQR

upper limit = +

Log Transformation

④ Feature Scaling
Maintaining all features at standard scale.

StandardScalar( )
↑ At any standard value.

MinMaxScalar( )
↑ b/w 0 and 1.

Sklearn.preprocessing

⑤ Feature Selection and Feature Engineering

① Selection
If correlation b/w two columns is 1 drop one.
data.drop(" ")
.corr( )

② Engineering
Make feature from previous features.

③ Dimensionality Reduction
Same sort of columns are combined and merged in one column.

PCA
.fit( ) ← train data.

⑥ Handling Imbalanced Dataset
classes have imbalanced no. of dataset.
e.g: class 1 → 5 rows
class 2 → 50 row

① Oversampling (SMOTE)
add more class row.

② Undersampling
reduce class rows.

③ class weighting
consider the same dataset and process it on same scale.

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CamScanner